Time Series Analysis

Part 1: Lecture 3b

(S)ARIMA Examples

Outline

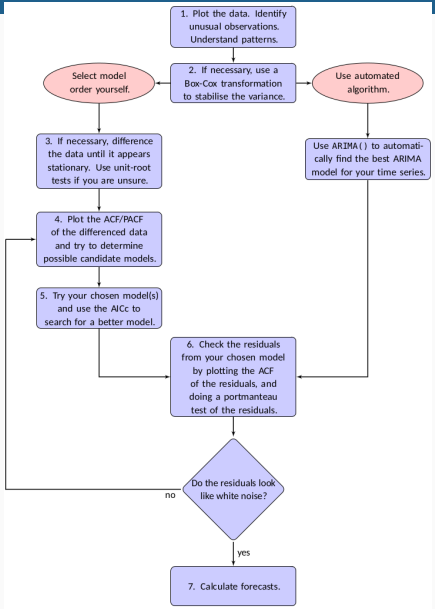
1 Modelling procedure

2 Seasonally adjusted electrical equipment

3 European quarterly retail trade

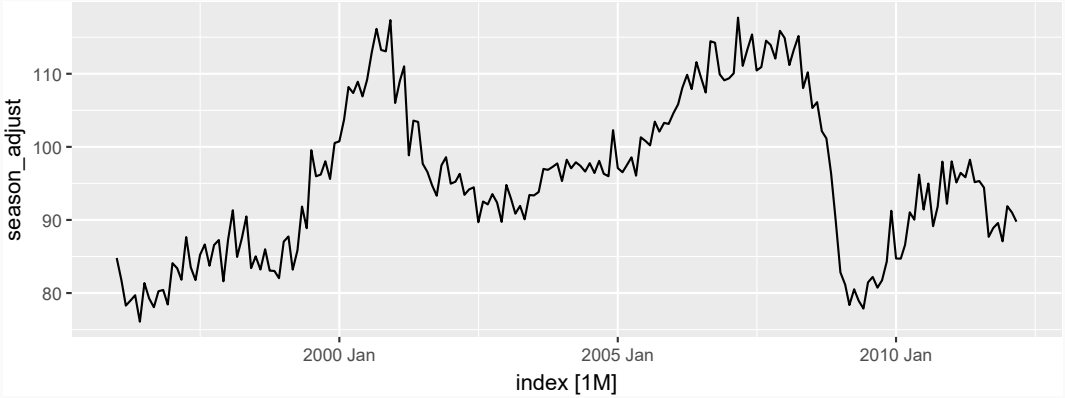
4 Cortecosteroid drug sales

Modelling procedure



Seasonally adjusted electrical equipment

```
elec_equip <- as_tsibble(fpp2::elecequip)
elec_dcmp <- elec_equip %>%
  model(STL(value ~ season(window="periodic"))) %>%
  components() %>% select(-.model) %>% as_tsibble()
elec_dcmp %>%autoplot(season_adjust)
```



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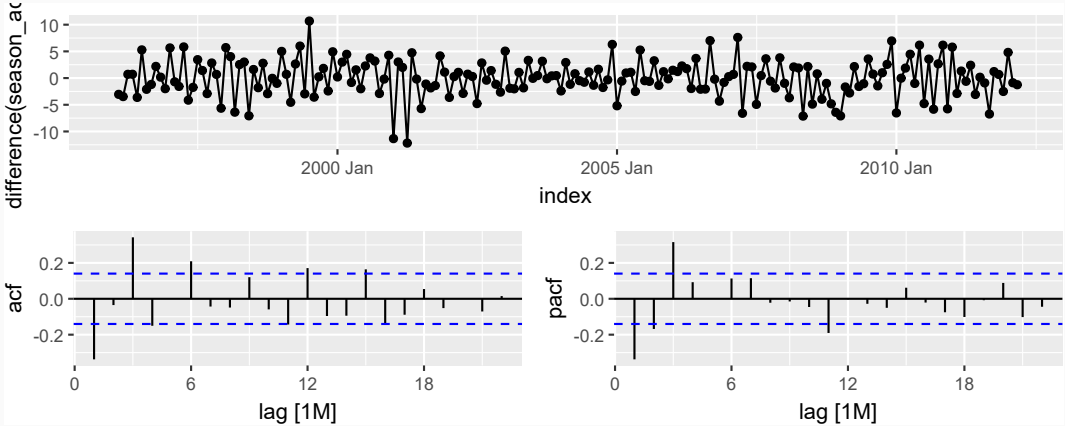
4 Cortecosteroid drug sales

Seasonally adjusted electrical equipment

1. Time plot shows sudden changes, particularly big drop in 2008/2009 due to global economic environment. Otherwise nothing unusual and no need for data adjustments.
2. No evidence of changing variance, so no Box-Cox transformation.
3. Data are clearly non-stationary, so we take first differences.

Seasonally adjusted electrical equipment

```
elec_dcmp %>%
  gg_tsdisplay(difference(season_adjust), plot_type='partial')
```



Seasonally adjusted electrical equipment

4. PACF is suggestive of AR(3). So initial candidate model is ARIMA(3,1,0). No other obvious candidates.
5. Fit ARIMA(3,1,0) model along with variations: ARIMA(4,1,0), ARIMA(2,1,0), ARIMA(3,1,1), etc. ARIMA(3,1,1) has smallest AICc value.

Seasonally adjusted electrical equipment

```
fit <- elec_dcmp %>%
  model(
    arima = ARIMA(season_adjust ~ pdq(3,1,1) + PDQ(0,0,0))
  )
report(fit)
```

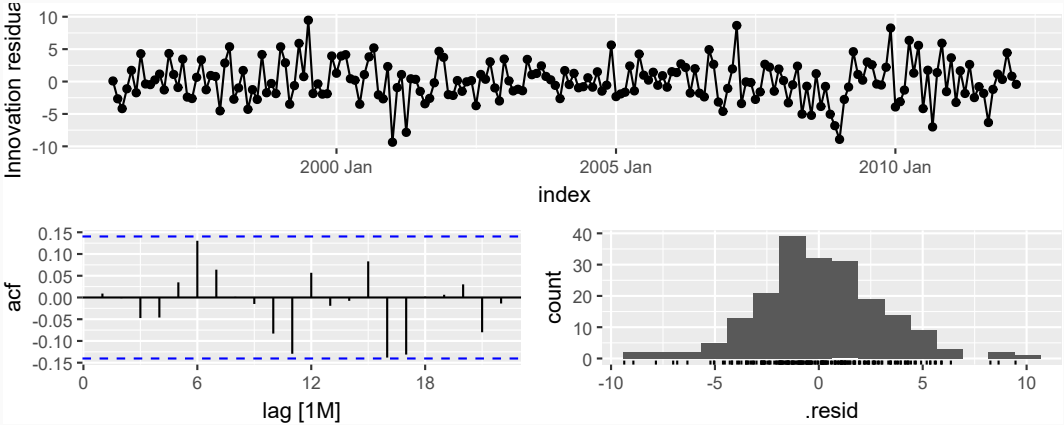
```
Series: season_adjust
Model: ARIMA(3,1,1)

Coefficients:
         ar1     ar2     ar3      ma1
      0.0044  0.0916  0.3698  -0.3921
s.e.  0.2201  0.0984  0.0669   0.2426

sigma^2 estimated as 9.577:  log likelihood=-492.7
AIC=995.4   AICc=995.7   BIC=1012
```

Seasonally adjusted electrical equipment

6 ACF plot of residuals from ARIMA(3,1,1) model look like white noise.

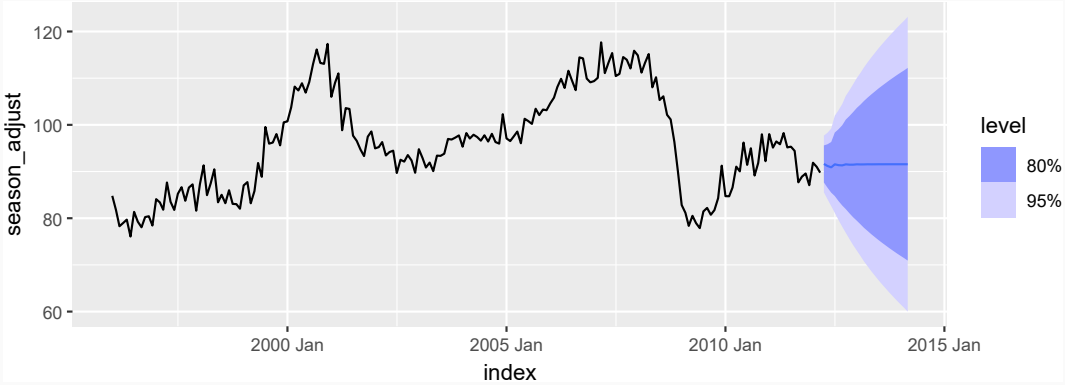


Seasonally adjusted electrical equipment

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>      <dbl>
1 arima     24.0      0.241
```

Seasonally adjusted electrical equipment

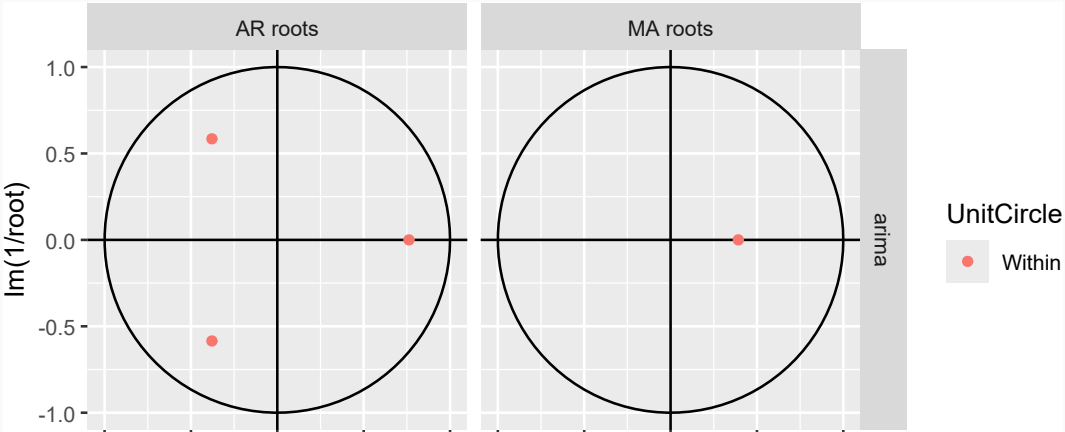
```
fit %>% forecast() %>% autoplot(elec_dcmp)
```



Roots checking

- The three red dots correspond to the roots of the polynomials $\phi(B)$ (left) and $\theta(B)$ (right)of the ARIMA(3,1,1) model.

```
gg_arma(fit)
```



Roots checking

- They are all inside the unit circle, as we would expect because R ensures the fitted model is both stationary and invertible.
- Any roots close to the unit circle may be numerically unstable, and the corresponding model will not be good for forecasting.
- The `ARIMA()` function will never return a model with inverse roots outside the unit circle.

Outline

1 Modelling procedure

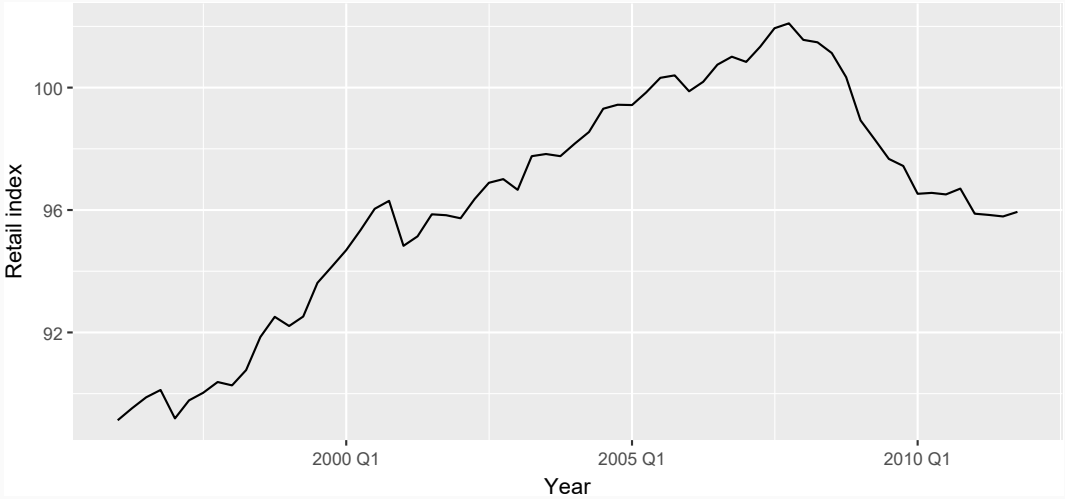
2 Seasonally adjusted electrical equipment

3 European quarterly retail trade

4 Cortecosteroid drug sales

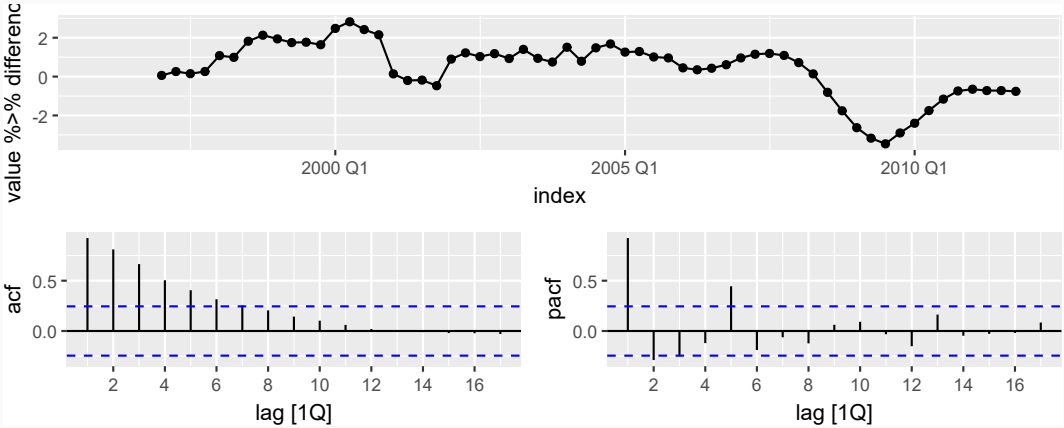
European quarterly retail trade

```
eu_retail %>% autoplot(value) +
  xlab("Year") + ylab("Retail index")
```



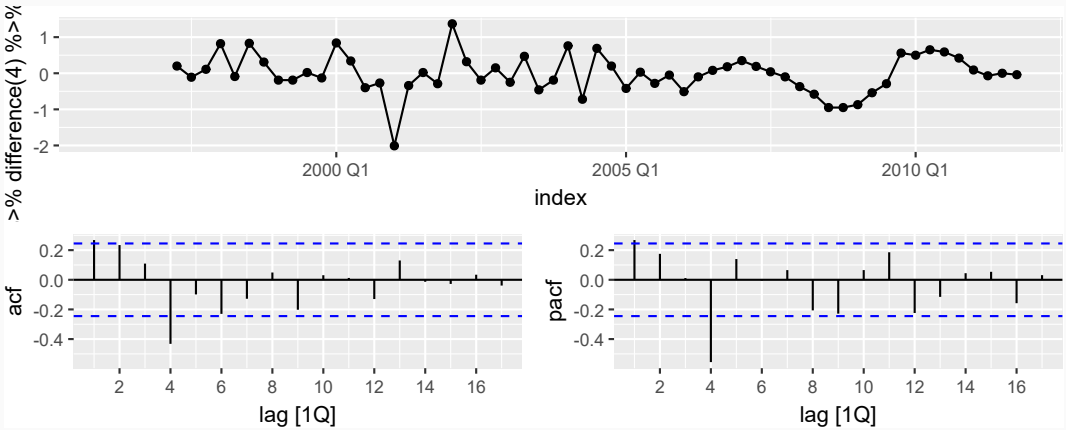
European quarterly retail trade

```
eu_retail %>% gg_tsdisplay(
  value %>% difference(4), plot_type='partial')
```



European quarterly retail trade

```
eu_retail %>% gg_tsdisplay(
  value %>% difference(4) %>% difference(1),plot_type='partial')
```

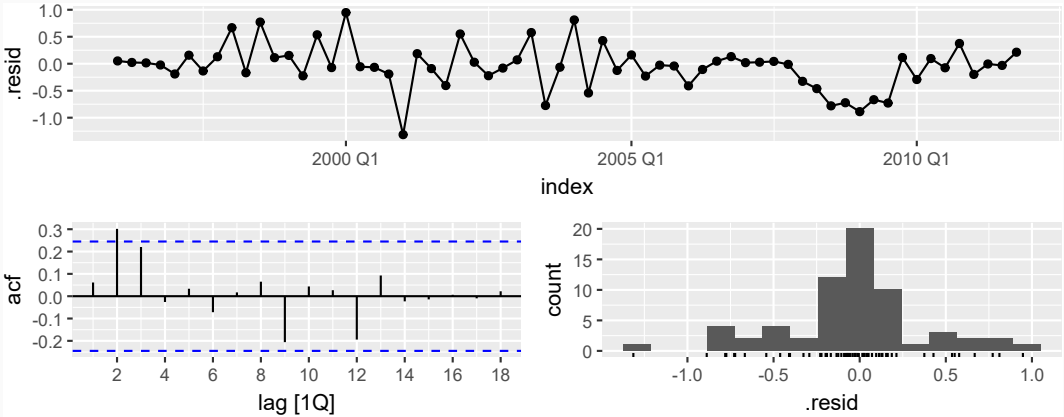


European quarterly retail trade

- $d = 1$ and $D = 1$ seems necessary.
- Significant spike at lag 1 in ACF suggests non-seasonal MA(1) component.
- Significant spike at lag 4 in ACF suggests seasonal MA(1) component.
- Initial candidate model: ARIMA(0,1,1)(0,1,1)$_4$.
- We could also have started with ARIMA(1,1,0)(1,1,0)$_4$.

European quarterly retail trade

```
fit <- eu_retail %>%
  model(arima = ARIMA(value ~ pdq(0,1,1) + PDQ(0,1,1)))
augment(fit) %>% gg_tsdisplay(.resid, plot_type = "hist")
```



European quarterly retail trade

```
augment(fit) %>%
  features(.resid, ljung_box, lag = 8, dof = 2)
```

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>     <dbl>     <dbl>
1 arima      10.7    0.0997
```

European quarterly retail trade

- ACF and PACF of residuals show significant spikes at lag 2, and maybe lag 3.
- AICc of ARIMA(0,1,1)$(0,1,1)_4$ model is 75.72
- AICc of ARIMA(0,1,2)$(0,1,1)_4$ model is 74.27.
- AICc of ARIMA(0,1,3)$(0,1,1)_4$ model is 68.39.
- AICc of ARIMA(0,1,4)$(0,1,1)_4$ model is 70.73.

European quarterly retail trade

```
fit <- eu_retail %>%
  model(
    arima013011 = ARIMA(value ~ pdq(0,1,3) + PDQ(0,1,1))
  )
report(fit)
```

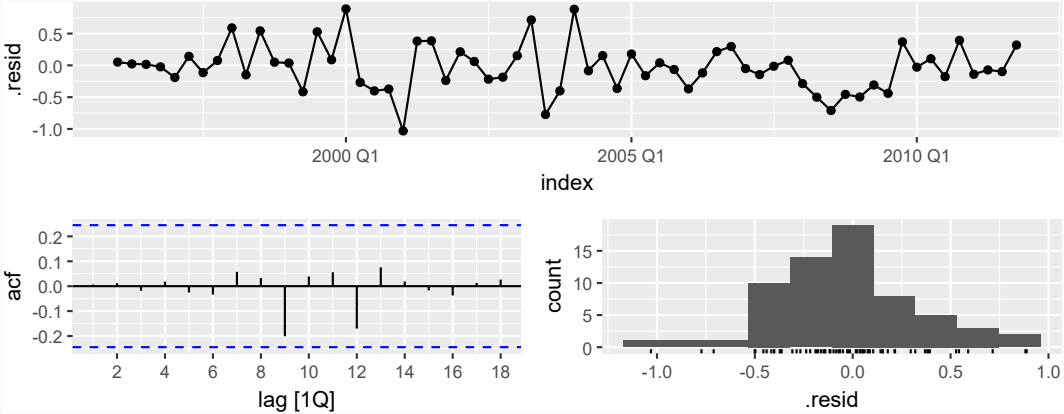
```
Series: value
Model: ARIMA(0,1,3)(0,1,1)[4]

Coefficients:
          ma1     ma2     ma3     sma1
      0.2630  0.3694  0.4200  -0.6636
s.e.  0.1237  0.1255  0.1294   0.1545

sigma^2 estimated as 0.156:  log likelihood=-28.63
AIC=67.26   AICc=68.39   BIC=77.65
```

European quarterly retail trade

```
augment(fit) %>%
  gg_tsdisplay(.resid, plot_type = "hist")
```



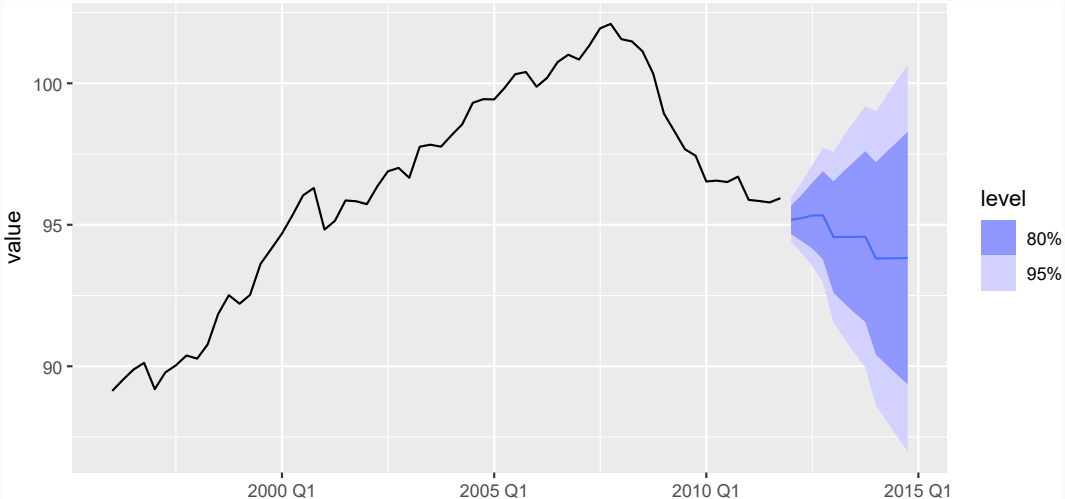
European quarterly retail trade

```
augment(fit) %>%
  features(.resid, ljung_box, lag = 8, dof = 4)
```

```
# A tibble: 1 x 3
  .model      lb_stat lb_pvalue
  <chr>         <dbl>     <dbl>
1 arima013011   0.511     0.972
```

European quarterly retail trade

```
fit %>% forecast(h = "3 years") %>%
  autoplot(eu_retail)
```



European quarterly retail trade

```
eu_retail %>% model(ARIMA(value)) %>% report()
```

```
Series: value
Model: ARIMA(0,1,3)(0,1,1)[4]

Coefficients:
          ma1     ma2     ma3     sma1
      0.2630  0.3694  0.4200  -0.6636
s.e.  0.1237  0.1255  0.1294   0.1545

sigma^2 estimated as 0.156:  log likelihood=-28.63
AIC=67.26   AICc=68.39   BIC=77.65
```

European quarterly retail trade

```
eu_retail %>% model(ARIMA(value, stepwise = FALSE,
  approximation = FALSE)) %>% report()
```

```
Series: value
Model: ARIMA(0,1,3)(0,1,1)[4]

Coefficients:
          ma1     ma2     ma3     sma1
      0.2630  0.3694  0.4200  -0.6636
s.e.  0.1237  0.1255  0.1294   0.1545

sigma^2 estimated as 0.156:  log likelihood=-28.63
AIC=67.26   AICc=68.39   BIC=77.65
```

Outline

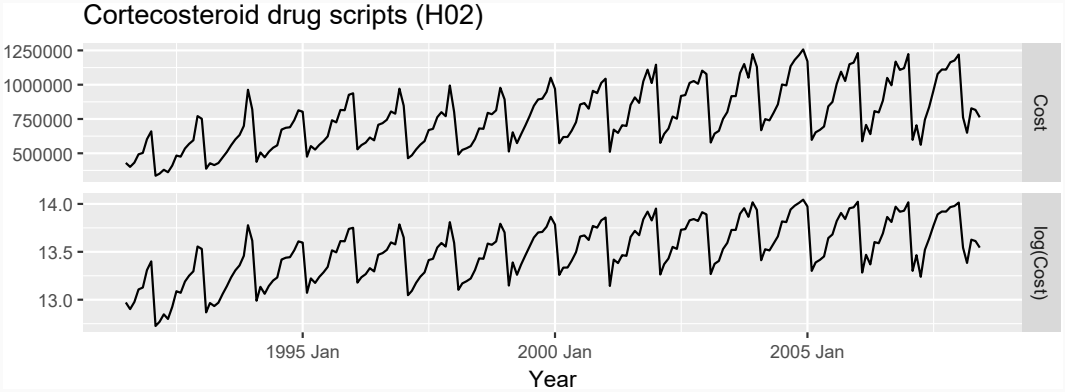
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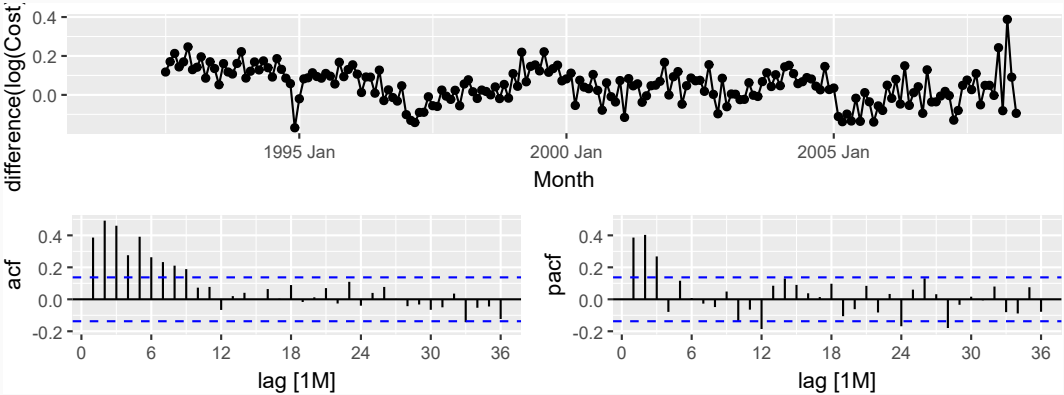
3 European quarterly retail trade

4 Cortecosteroid drug sales

Cortecosteroid drug sales



Cortecosteroid drug sales



Cortecosteroid drug sales

- Choose $D = 1$ and $d = 0$.
- Spikes in PACF at lags 12 and 24 suggest seasonal AR(2) term.
- Spikes in PACF suggests possible non-seasonal AR(3) term.
- Initial candidate model: ARIMA(3,0,0)(2,1,0)$_{12}$.

Cortecosteroid drug sales

.model	AICc
ARIMA(3,0,1)(0,1,2)[12]	-485.5
ARIMA(3,0,1)(1,1,1)[12]	-484.3
ARIMA(3,0,1)(0,1,1)[12]	-483.7
ARIMA(3,0,1)(2,1,0)[12]	-476.3
ARIMA(3,0,0)(2,1,0)[12]	-475.1
ARIMA(3,0,2)(2,1,0)[12]	-474.9
ARIMA(3,0,1)(1,1,0)[12]	-463.4

Cortecosteroid drug sales

```
fit <- h02 %>%
  model(best = ARIMA(log(Cost) ~ 0 + pdq(3,0,1) + PDQ(0,1,2)))
report(fit)
```

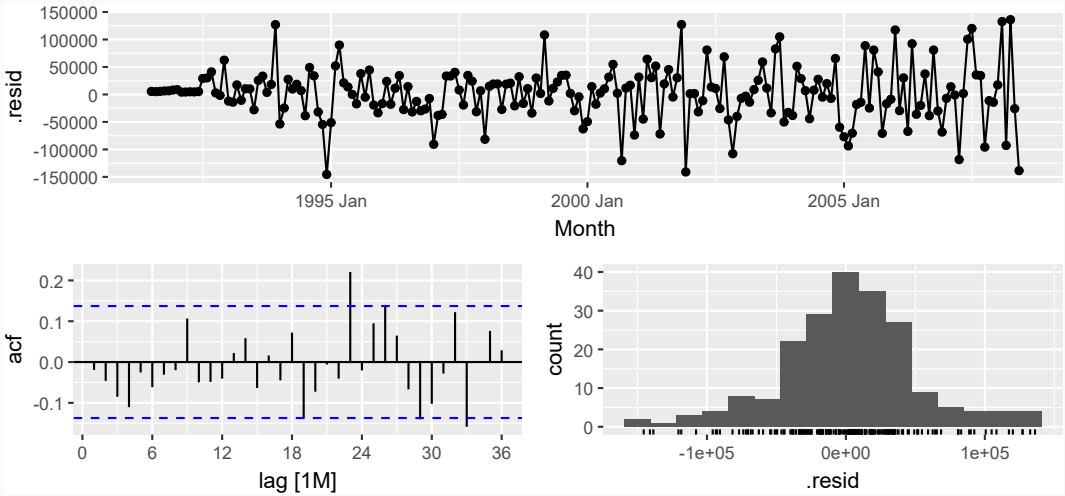
```
Series: Cost
Model: ARIMA(3,0,1)(0,1,2)[12]
Transformation: log(Cost)

Coefficients:
          ar1     ar2     ar3     ma1      sma1     sma2
      -0.1602  0.5481  0.5678  0.3826  -0.5222  -0.1769
s.e.   0.1636  0.0878  0.0942  0.1895   0.0861   0.0872

sigma^2 estimated as 0.004289:  log likelihood=250.1
AIC=-486.1   AICc=-485.5   BIC=-463.3
```

Cortecosteroid drug sales

```
augment(fit) %>%
  gg_tsdisplay(.resid, lag_max=36, plot_type = "hist")
```



Cortecosteroid drug sales

```
augment(fit) %>%
  features(.resid, ljung_box, lag = 36, dof = 6)
```

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>      <dbl>
1 best      58.0   0.00161
```

Cortecosteroid drug sales

```
fit <- h02 %>% model(auto = ARIMA(log(Cost)))
report(fit)
```

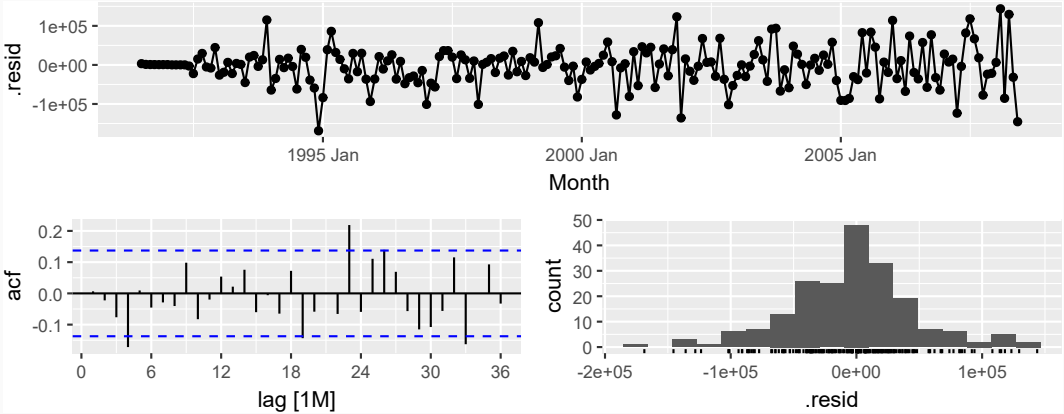
```
Series: Cost
Model: ARIMA(2,1,0)(0,1,1)[12]
Transformation: log(Cost)

Coefficients:
          ar1      ar2     sma1
      -0.8491  -0.4207  -0.6401
s.e.   0.0712   0.0714   0.0694

sigma^2 estimated as 0.004399:  log likelihood=245.4
AIC=-482.8   AICc=-482.6   BIC=-469.8
```

Cortecosteroid drug sales

```
augment(fit) %>%
  gg_tsdisplay(.resid, lag_max = 36, plot_type = "hist")
```



Cortecosteroid drug sales

```
augment(fit) %>%
  features(.resid, ljung_box, lag = 36, dof = 5)
```

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>     <dbl>
1 auto      63.8  0.000472
```

Cortecosteroid drug sales

```
fit <- h02 %>%
  model(best = ARIMA(log(Cost), stepwise = FALSE,
                     approximation = FALSE,
                     order_constraint = p + q + P + Q <= 9))
report(fit)
```

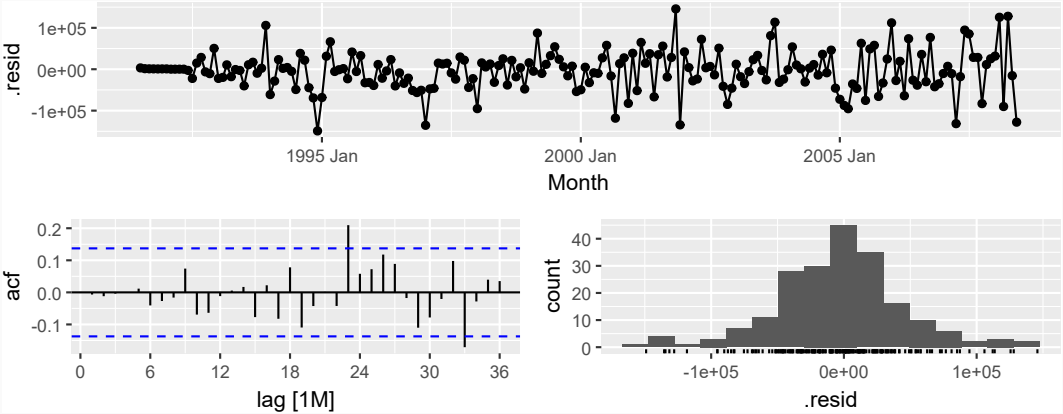
```
Series: Cost
Model: ARIMA(4,1,1)(2,1,2)[12]
Transformation: log(Cost)

Coefficients:
          ar1     ar2     ar3      ar4      ma1    sar1
      -0.0426  0.2097  0.2016  -0.2273  -0.7423  0.6213
s.e.   0.2168  0.1814  0.1144   0.0810   0.2075  0.2421
         sar2     sma1    sma2
      -0.3832  -1.2018  0.4958
s.e.   0.1185   0.2492  0.2136

sigma^2 estimated as 0.004061:  log likelihood=254.3
AIC=-488.6   AICc=-487.4   BIC=-456.1
```

Cortecosteroid drug sales

```
augment(fit) %>%
  gg_tsdisplay(.resid, lag_max = 36, plot_type = "hist")
```



Cortecosteroid drug sales

```
augment(fit) %>%
  features(.resid, ljung_box, lag = 36, dof = 9)
```

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>     <dbl>
1 best      44.1    0.0203
```

Cortecosteroid drug sales

Training data: July 1991 to June 2006

Test data: July 2006–June 2008

```
fit <- h02 %>%
  filter_index(~ "2006 Jun") %>%
  model(
    ARIMA(log(Cost) ~ pdq(3, 0, 0) + PDQ(2, 1, 0)),
    ARIMA(log(Cost) ~ pdq(3, 0, 1) + PDQ(2, 1, 0)),
    ARIMA(log(Cost) ~ pdq(3, 0, 2) + PDQ(2, 1, 0)),
    ARIMA(log(Cost) ~ pdq(3, 0, 1) + PDQ(1, 1, 0))
    # ... #
  )

fit %>%
  forecast(h = "2 years") %>%
  accuracy(h02 %>% filter_index("2006 Jul" ~ .))
```

```
# A tibble: 4 x 10
  .model     .type      ME   RMSE    MAE     MPE   MAPE  MASE
```

Cortecosteroid drug sales

Cortecosteroid drug sales

.model	RMSE
ARIMA(3,0,1)(1,1,1)[12]	61878
ARIMA(3,0,1)(0,1,2)[12]	62142
ARIMA(2,1,4)(0,1,1)[12]	62708
ARIMA(2,1,3)(0,1,1)[12]	62856
ARIMA(3,0,1)(0,1,1)[12]	62947
ARIMA(3,0,2)(0,1,1)[12]	62968
ARIMA(4,1,1)(2,1,2)[12]	63114
ARIMA(3,0,3)(0,1,1)[12]	63256
ARIMA(2,1,5)(0,1,1)[12]	63610
ARIMA(3,0,2)(2,1,0)[12]	65146
ARIMA(3,0,1)(2,1,0)[12]	65270
ARIMA(3,0,1)(1,1,0)[12]	66644
ARIMA(3,0,0)(2,1,0)[12]	66816

Cortecosteroid drug sales

- Models with lowest AICc values tend to give slightly better results than the other models.
- AICc comparisons must have the same orders of differencing. But RMSE test set comparisons can involve any models.
- Use the best model available, even if it does not pass all tests.

Cortecosteroid drug sales

```
fit <- h02 %>%
  model(ARIMA(Cost ~ 0 + pdq(3,0,1) + PDQ(1,1,1)))
fit %>% forecast %>% autoplot(h02) +
  ylab("H02 Expenditure ($AUD)") + xlab("Year")
```

